

Python Tasks:

Integer (int) - 10 if-else Tasks

1. Check if a number is odd or even.
 2. Determine if a number is a multiple of 3 and 5.
 3. Check if a number is positive, negative, or zero.
 4. Find the largest among three integers.
 5. Check if a number is a prime number.
 6. Determine if a number is within the range 100–999.
 7. Check if the sum of two integers is greater than 50.
 8. If a number is divisible by 4 but not by 8, print "Special".
 9. Compare two numbers and print "Equal", "Greater", or "Smaller".
 10. Classify age: Child (0–12), Teen (13–19), Adult (20–59), Senior (60+).
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Float (float) - 10 if-else Tasks

1. Check if a float number is positive.
 2. Determine if a float is within 0.0 to 1.0 (inclusive).
 3. If temperature > 38.0, print "Fever"; else "Normal".
 4. Compare two float numbers and print the larger one.
 5. Check if BMI (float) is underweight (<18.5), normal, overweight, or obese.
 6. If bank balance > 1000.50, print "Sufficient"; else "Low Balance".
 7. Round the float to 1 decimal and check if it's > 5.5.
 8. Check if float * 2 is greater than 10.
 9. Determine if a float value is exactly halfway between two integers.
 10. Check if a float value is close to 3.14 within a tolerance of 0.01.
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String (str) - 10 if-else Tasks

1. Check if a string is a palindrome.
2. Verify if a string contains only alphabets.
3. Check if a string contains both "Python" and "AI".
4. If string length is greater than 10, print "Long"; else "Short".
5. Determine if a string starts with a capital letter.
6. Compare two strings and print if they are the same or not.
7. Check if a string ends with ".com".
8. Count number of vowels and print "Vowel-rich" if more than 3.
9. If a string contains any digits, print "Contains numbers".
10. Check if a password has length ≥ 8 and no spaces.

List (`list`) - 10 if-else Tasks

1. Check if a list is empty.
 2. Determine if the first and last elements are the same.
 3. If a list contains more than 5 items, print "Long List".
 4. Check if a specific value (e.g. 7) exists in the list.
 5. Compare lengths of two lists and print which is longer.
 6. If sum of all numbers in list > 100, print "High Sum".
 7. Check if all elements in a list are even.
 8. Check if list is sorted in ascending order.
 9. If list has any duplicate values, print "Has Duplicates".
 10. If list contains only positive numbers, print "All Positive".
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Dictionary (`dict`) - 10 if-else Tasks

1. Check if a key 'name' exists in the dictionary.
 2. If 'marks' key exists and value ≥ 40 , print "Pass".
 3. Check if a dictionary is empty.
 4. Compare values of two keys and print which is greater.
 5. If all values are > 50, print "Above Average".
 6. Check if value of 'age' is within 18–60.
 7. If 'active' is `True`, print "User Active".
 8. If 'price' is less than 'discount_price', print "Invalid Discount".
 9. Count how many keys have values as integers.
 10. Check if nested dictionary key 'user' -> 'admin' exists and is `True`.
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Boolean (`bool`) - 10 if-else Tasks

1. If a boolean variable is `True`, print "Yes"; else "No".
2. Check if both `a` and `b` are `True`.
3. If either `a` or `b` is `True`, print "At least one is True".
4. If `is_logged_in` is `False`, print "Please login".
5. If `is_admin` and `is_verified` are both `True`, print "Access granted".
6. If `has_permission` is not `True`, print "Restricted".
7. Check if two boolean flags differ (XOR logic).
8. Toggle a boolean: if `True` make it `False`, else `True`.
9. If a boolean variable is missing or `None`, print "Invalid".
10. Use a boolean to determine visibility of a UI element.

